

Electromagnetic Theory (EE3005)

Date: November 01, 2025

Course Instructor(s)

1. Dr. Afzal Ahmed (Course Moderator)
2. Mohsin Yousuf

Sessional-II Exam

Total Time (Hrs): 1
Total Marks: 40
Total Questions: 2

Roll No

Section

Student Signature

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1. Attempt all questions and remember to solve parts of the same question together.
2. Final answers should be correct up to two decimal places with proper SI units.
3. Show all the steps with the help of diagrams and equations.

CLO # 02: Analyze electrostatic fields and/or its properties governed by Coulomb's / Gauss's law for a given charge distribution in free space and / or dielectrics.

Q1:

[20 marks]

(a) Given the field,

[10]

$$\mathbf{D} = 6\rho \sin \frac{1}{2}\varphi \mathbf{a}_\rho + 1.5\rho \cos \frac{1}{2}\varphi \mathbf{a}_\varphi \text{ [C/m}^2\text{]}$$

Examine both sides of the *divergence theorem* for the region bounded by $\rho = 2$, $\varphi = 0$, $\varphi = \pi$, $z = 0$, and $z = 5$.

Hint:

$$\nabla \cdot \mathbf{D} = \frac{1}{\rho} \frac{\partial}{\partial \rho}(\rho D_\rho) + \frac{1}{\rho} \frac{\partial D_\varphi}{\partial \varphi} + \frac{\partial D_z}{\partial z}$$

- (b) Four point-charges $Q_1 = +q$, $Q_2 = -q$, $Q_3 = -q$, and $Q_4 = +q$ are located at $(0, 0)$, $(a, 0)$, $(0, a)$, and (a, a) respectively in yz -plane as shown in Figure 1, where $q = 0.8 \text{ nC}$ and $a = 4 \text{ cm}$. Find the total potential energy stored in this quadrupole.

[10]

Hint: $W_E = \frac{1}{2} \sum Q_i V_i$

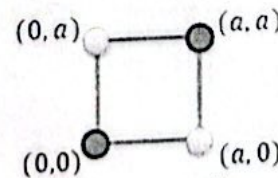


Figure 1. Quadrupole Charges

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[20 marks]

[9]

Q2:

- (a) Current charge density is defined as $\mathbf{J} = -10^6 z^{5/2} \mathbf{a}_z$ [A/m²] in CCS within the region $0 \leq \rho \leq 30 \mu\text{m}$ and is zero ($\mathbf{J} = 0$), elsewhere. Answer the following illustrating the linear relationship between *current density* and *charge density* as well as *velocity*:

- (i) Find the total current ' I ' crossing the surface $z = 0.1$ m in the $\mathbf{a}_z$ direction.
- (ii) If the charge velocity ' $\mathbf{v}$ ' is 3×10^6 m/s at $z = 0.1$ m, find ρ_v there.
- (iii) If the volume charge density ' ρ_v ' at $z = 0.15$ m is -3000 [C/m³], find the charge velocity there.

[11]

- (b) Two extensive homogeneous isotropic dielectrics meet on plane $z = 0$. Let Region 1 ($z \geq 0$) have relative permittivity $\epsilon_{r1} = 3$, while Region 2 ($z \leq 0$) is characterized by $\epsilon_{r2} = 5$. A uniform electric field $\mathbf{E}_1 = 3\mathbf{a}_x - 2\mathbf{a}_y + 5\mathbf{a}_z$ V/m exists for $z \geq 0$.
Extrapolate (i) E_{N1} , (ii) E_{t1} , (iii) E_{t1} , (iv) E_1 , (v) E_{t2} , (vi) E_{N2} , (vii) E_2 , (viii) P_1 , (ix) P_2 , (x) D_1 and (xi) D_2 .

Hint Part (b): $z = \text{const.}$ is an xy -plane, ' z ' component of any vector field corresponds to the *normal* component, whereas, ' x ' and ' y ' components are *tangential*. Use boundary conditions for dielectric-to-dielectric interface.